

Mobile App Development with Flutter

Course Focus: Cross-Platform Mastery for iOS & Android

Level: Beginner to Intermediate

Course Description

Why build two apps when you can build one? This course is designed to turn you into a versatile mobile developer using Google's Flutter framework.^[1] We start with the Dart language and move quickly into building beautiful, high-performance UIs. We won't just make "toy" apps; we'll focus on state management and API integration used in real-world tech companies.^[1]

What You'll Learn

- The **Dart** programming language (Null safety, Async/Await).
- The "Everything is a Widget" philosophy.
- State Management (we'll focus on **Riverpod** or **BLoC**).^[1]
- Connecting to **Firebase** and external REST APIs.

Weekly Breakdown

- **Week 1: Hello Dart** – Variables, Functions, OOP in Dart, and Null Safety.
- **Week 2: Flutter Basics** – Layouts (Rows, Columns, Stacks) and Basic Widgets.
- **Week 3: Navigation & Routing** – Moving between screens and passing data.^[2]
- **Week 4: User Input** – Forms, TextFields, and Validation logic.
- **Week 5: State Management Deep Dive** – Why `setState` isn't enough. Learning **Riverpod** for scalable apps.^[1]
- **Week 6: Networking** – Fetching data from APIs using the `http` or `dio` package.
- **Week 7: Persistence & Firebase** – Local storage (`SharedPreferences`) and Cloud Firestore for real-time data.
- **Week 8: Polishing & Publishing** – Custom animations, App icons, and the App Store/Play Store release process.

Final Project

A **Personal Finance Tracker** with data visualization (charts), cloud sync, and offline support.